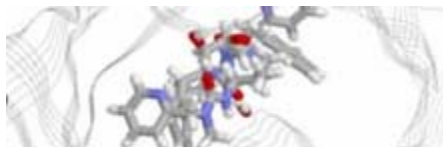


I571 Chemical Information Technology Information Pack for Participants



Indiana University School of Informatics Certificate in Chemical Informatics

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informatics

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Essential Information

I571 CHEMICAL INFORMATION TECHNOLOGY is a 3 credit hour graduate class aimed at introducing students to the fundamental informatics techniques for chemistry-related disciplines. The class will cover a wide range of topics, including representation and use of chemical structure information, computer-aided drug design, 3D visualization and computation, and handling of large volumes of chemical information.

Course Tutors

The course is being taught by Dr. David Wild, who is your first contact for all matters relating to the course. You can contact him by email (djwild@indiana.edu) or phone (734-769-5237). There will also be course tutors at the Bloomington and Indianapolis sites, to assist with local matters there. They are Dr. Gary Wiggins for Bloomington (wiggins@indiana.edu, phone 812-856-1086) and Dr. Kelsey Forsythe for Indianapolis (kforsyth@iupui.edu, phone 317-278-2202).

		
David Wild IU Bloomington	Gary Wiggins IU Bloomington	Kelsey Forsythe IUPUI Indianapolis

Course Times & Dates

I571 be held on Mondays and Wednesdays 9.30-10.45am *Indiana (Eastern Standard) Time*. Since Indiana does not use Daylight Savings, this time will be the same as Central Time until October 30th, and then will be the same as Eastern Time. To help you figure out the times for you take a look at the conversion table below. The first class will be held on **August 29th** and the last on **December 7th**. Note that, in line with Indiana University tradition, **there will be a class on Labor Day (Sept. 5th)**.

Time Zone	Before October 30	After October 30th
Eastern	10.30-11.45	9.30-10.45
Central	9.30-10.45	8.30-9.45
Mountain	8.30-9.45	7.30-8.45
Pacific	7.30-8.45	6.30-7.45
GMT / UT	2.30-3.45pm	2.30-3.45pm

Course Grading

The course will be graded A-C (with A and B as passing grades), based on twelve short assignments given at the end of classes (60% of grade, 5% each) and a more substantial main project at the end (40% of the grade). The main project will include options to suit a variety of tastes and skills (from essay questions to programming assignments).

Recommended textbooks

1. Andrew Leach & Valerie Gillet, *An Introduction to Chemoinformatics*. Kluwer Academic Publishers, Dordrecht, NL, 2003.
2. Johann Gasteiger & Thomas Engel (eds.), *Chemoinformatics: A Textbook*. Wiley-VCH, Weinheim, 2003

3. Jürgen Bajorath, *Chemoinformatics: Concepts, Methods, and Tools for Drug Discovery*. Humana Press, Totowa, New Jersey, 2004
4. David Wild, *Getting Started in Chemoinformatics*. Available in PDF format on Oncourse

How to participate in the class using Raindance (distance students only)

To enable participants to join in from a distance, we are using a service from Raindance (www.raindance.com). This service incorporates the *Meeting Edition* software for sharing of slides, applications and for participant interaction, and telephone audio conferencing for lecture audio and discussion.

Installing the software before the first class

You can choose either the *full* or *light* versions of the software to use. If you are using a Mac, Linux or a non-Windows PC (Windows 98SE or later), or you don't have privileges to install new software on your machine, you can just choose the "light" version when you join the lecture (see below), and you don't need to do anything at this stage. If you are able to install the full version, it will give you more functionality (such as being able to share your video with a webcam). To install it, go to <http://indianaedu.on.raindance.com>, and click on the "Pre-meeting Install" tab, click the "Install Now" button, and wait for the install to start. You will need to click "Yes" to accept the certificate. When the install is complete and you get the "Buy, Try, Add" screen, you can just exit the application until you are ready to join the first lecture.

Participating in a lecture

The Raindance conference will normally be opened five minutes before the start of each lecture, although for the first lecture we will open it 15 minutes early.

If you are using the full version of the software (see above), open the Raindance application which was installed on your computer, and click on "Join a meeting" (the small link under Buy, Try, Add). When prompted, enter your name, email and conference ID **1533687**. If you are using the light version of the software, open your browser and go to <http://indianaedu.on.raindance.com>. You will be asked to enter your name, email address, company and the conference ID, which is **1533687**. Choose "Light Version", then click on the *Join Now* button. After a short time the Java application will open.

If you join before we open the conference, the software will just wait until it is opened. Once the meeting has begun, you will be prompted to enter a phone number where Raindance can call you for the audio portion. If you prefer, you can cancel and dial the toll-free number 1-888-272-7337 yourself (and enter the conference ID above). Either way, you will not have to pay phone charges. If you are joining from outside the US, you can also call the regular (toll) number 1-303-928-2688.

If you experience technical difficulties during a class, you should let me know either by flagging that you have a problem in the Meeting Edition software (you will be shown how to do this in the first class), by alerting us to the fact on the phone teleconference, or by calling my cell phone at 734-709-0753. You can also access Raindance technical support immediately by pressing **00** during a phone conference, or by calling 1-877-549-2051.

What to do if you miss a class

If you miss a class, please let me know at djwild@indiana.edu. All of our classes will be recorded, and the replay files will be made available either in Oncourse or on a separate website. To play back the files, you will need to download and install the Raindance playback software, available from <https://indianaedu.on.raindance.com/confmgr/installs/RaindancePlayer.exe>.

Accessing the Oncourse repository

We shall be using *Oncourse*, Indiana University's Course Management System, as a repository for documents, coursework, slide presentations, web links and other materials relating to the course. To access Oncourse you will need an Indiana University username and password, which you should be assigned automatically after registering for this course. To access Oncourse, go to oncourse.iu.edu, and click on the log in button at the top right. After doing this, click on the **BL INFO I571 29446** link to go to the site for this class (not BL INFO I571 29447 or IN INFO I571 28551). If you do not see this link in Oncourse, let me know

and I will look into it. If you have trouble obtaining a username and password, let me know and I will give you a temporary guest login.

Course Schedule

There may be some minor changes to this schedule, but it should stay basically intact.

August 29 th	Introduction to the world of chemical informatics <i>Overview of the class; defining chemical informatics; chemical and bioinformatics; chemical informatics and the pharmaceutical industry; example applications; a glimpse of the future of chemical informatics</i>
August 31 st	Representing 2D structures I <i>Kinds of 2D structure representation; atom lookup and connection tables; graph theory; SMILES; SD files; representation nuances; descriptors</i>
Sept. 5 th (Labor Day)	Representing 2D structures II <i>Canonicalizing SMILES; representing reactions; representing generic structures; fingerprints; measuring chemical similarity; XML, CML, InChI and other web-oriented representation systems</i>
Sept. 7 th	2D chemical database applications <i>Types of searching; substructure searching with SMARTS; similarity searching with fingerprints; demonstrations of searching systems</i>
Sept. 12 th	Advanced 2D descriptors <i>Kinds of descriptor; "mathematical" and topological indices; biological descriptors and their application in ADME/Tox; biological properties; property prediction software</i>
Sept. 14 th	Advanced 2D methods <i>2D QSAR; cluster analysis; diversity analysis</i>
Sept. 19 th	Representing 3D structures <i>Sources of 3D information; experimental 3D databases; conformational flexibility; distance matrices; estimation of 3D structure; conformational search and minimization; 3D descriptors and fingerprint; representation of proteins</i>
Sept. 21 st	3D visualization & computation <i>History of 3D visualization; free visualization and computation tools; 3D QSAR and models of activity; docking & virtual screening</i>
Sept. 26 th	Chemical informatics, bioinformatics, genomics and proteomics <i>The developing relationship between these fields</i>
Sept. 28 th	Practical high throughput screening analysis <i>A walkthrough of high throughput screening data analysis (Dr. Jack Bikker, Pfizer Global Research & Development)</i>
October 3 rd	Practical molecular modeling <i>A walkthrough of the application of molecular modeling in drug design (Dr. Tom Doman, Eli Lilly)</i>
October 5 th	Hot topics in the Pharmaceutical Industry <i>A discussion of current hot topics, including the intelligent use of large volumes of chemical and related information (Dr. Tom Doman, Eli Lilly)</i>
October 10 th	Laboratory information management systems <i>Workflows in the laboratory; LIMS and SDMS architectures (Dr. Doug Perry, IUPUI)</i>
October 12 th	Laboratory information management systems II <i>Current state of the art and issues (Dr. Doug Perry, IUPUI)</i>
October 17 th	Commercial vendors & databases I <i>Systems for browsing databases (STN, SciFinder Scholar, Crossfire Commander); Commercial databases available (CAS, Beilstein, Cambridge Structural Dataset); Spectral Databases; Open Access issues (Dr. Gary Wiggins, Indiana University)</i>
October 19 th	Commercial vendors & databases II <i>Product demonstrations (Dr. Gary Wiggins, Indiana University)</i>
October 24 th	Electronic laboratory notebooks <i>Kinds of E-lab notebooks; authentication issues; adoption in the pharmaceutical industry; e-lab notebook functionality (Dr. Doug Perry, IUPUI)</i>
October 26 th	Chemical informatics software development I <i>Languages and programming toolkits; client/server software and web-based interfaces; the semantic web</i>
October 31 st	Chemical & material science informatics in the process chemical industry <i>Characteristics of the industry; software development approaches; architectures (Dr. Bruce Wilson, Dow Chemical)</i>
Nov. 2 nd	Chemical informatics software development II <i>High performance computing; designing software for scientists</i>
Nov. 7 th	Artificial Intelligence in chemical informatics <i>Foundations of AI; neural networks; recursive partitioning; support vector machines; genetic algorithms; intelligent agents</i>
Nov. 9 th	Computational theoretical chemistry <i>Molecular mechanics and quantum mechanics (Dr. Tom Doman, Eli Lilly)</i>
Nov. 14 th	Computational models for ADME/Tox <i>The application of predictive models to pharmacology and toxicity testing (Dr. Sean Ekins, GeneGo)</i>
Nov. 16 th	Overview of chemical informatics research <i>An overview of the state of the art in chemical informatics research</i>

Nov. 21 st	Literature Review <i>A discussion of current chemical informatics literature</i>
Nov. 23 rd	Generic structures and searching <i>A detailed study of the representation and use of generic structures for patent and combinatorial chemistry applications (Dr. John Barnard, Barnard Chemical Information)</i>
Nov. 28 th	Emerging web service technologies for chemical informatics <i>A detailed look at how recent web developments such as the Semantic Web, web services, XML and RSS are likely to impact chemical informatics</i>
Nov. 30 th	Careers in chemical informatics <i>A discussion of available careers in chemical informatics</i>
Dec. 5 th	Class review, questions and answers <i>A panel Q&A session</i>
Dec. 7 th	Class review, questions and answers <i>Review of the class, general questions and answers</i>

Course Software

Below is a list of the main software packages that we shall be exploring in the I571 class. Some will be used for coursework, and some will be used only for demonstration. You can find links to many of the packages listed in Oncourse (oncourse.iu.edu).

Software available for unrestricted use (open source or freeware)

<i>ArgusLab</i>	3D visualization and molecular modeling www.arguslab.com
<i>ChemTK Lite</i>	2D dataset organization and analysis (limited functionality and only 250 structures) www.sageinformatics.com
<i>JMol</i>	Visualization of 3D chemical structures jmol.sourceforge.net
<i>PuTTY</i>	SSH client for accessing Unix machines http://www.chiark.greenend.org.uk/~sgtatham/putty/
<i>Taverna</i>	Workflow and pipelining tool taverna.sourceforge.net

Software available for class use only (licensed to Indiana University)

<i>BCI Toolkit</i>	C/C++ toolkit for fingerprint manipulation, cluster analysis and handling generic structures
<i>ChemAxon Toolkits</i>	Java applications for property calculation and 2D structure visualization
<i>Daylight Toolkit</i>	Manipulation of 2D chemical structure information
<i>Molinspiration Toolkits</i>	Java applications for database searching, property calculation and virtual screening
<i>OpenEye Toolkit</i>	C/C++ toolkit for 2D and 3D structure manipulation

Software available for use on request (licensed to Indiana University)

<i>ChemTK</i>	2D dataset organization and analysis, full version
<i>Sentient</i>	IO-Informatics knowledge integration package
<i>Spotfire DecisionSite</i>	Statistical analysis and visualization
<i>SYBYL</i>	Tripos molecular modeling package